

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario based

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Scenario Based	Assessment:	Assessment Tool 1– Scenario Based
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	To apply the statistical inferences such as testing of hypothesis and point estimates for the given data		
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Code of Conduct

I Deva Dharshini(192472350) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Deva Dharshini

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. A software company claims that its new application has an average response time of less than 2 seconds. A hypothesis test gives a p-value of 0.004. At $\alpha = 0.01$, determine whether there is sufficient evidence to support the company's claim.

Scenario

A software company claims that its newly developed application has an average response time of less than 2 seconds. To verify this claim, the company performs a hypothesis test using sample data collected from the application. The statistical analysis produces a p-value of 0.004. The significance level selected for the test is $\alpha = 0.01$. The objective is to determine whether there is sufficient statistical evidence to support the company's claim that the average response time is less than 2 seconds.

Answer

In this scenario, the company is making a claim about the population mean response time of its application. Let μ represent the true average response time of the application.

The company claims that the average response time is less than 2 seconds. Therefore, the statistical hypotheses can be formulated as follows:

Step 1: Formulation of Hypotheses

Null Hypothesis (H_0):

The null hypothesis represents the situation in which the company's claim is not supported. It assumes that the average response time is at least 2 seconds.

Alternative Hypothesis (H_1):

The alternative hypothesis represents the company's claim that the application's average response time is less than 2 seconds.

Since the alternative hypothesis contains the symbol "<", this is a left-tailed hypothesis test.

Step 2: Given Information

The following values are provided:

- p-value = 0.004
- Significance level, $\alpha = 0.01$

The significance level represents the maximum probability of making a Type I error that the researchers are willing to accept. A Type I error occurs when the null hypothesis is rejected even though it is actually true.

In this case, $\alpha = 0.01$ means that the company is using a strict 1% significance level. Therefore, strong evidence is required before rejecting the null hypothesis.

Step 3: Comparison of p-value and Significance Level

The p-value approach to hypothesis testing follows the rule:

- If $p\text{-value} \leq \alpha$, reject the null hypothesis.
- If $p\text{-value} > \alpha$, fail to reject the null hypothesis.

Now, compare the given values:

The p-value is smaller than the selected significance level.

Therefore, we reject the null hypothesis.

Step 4: Interpretation of the p-value

The p-value of 0.004 indicates that, assuming the null hypothesis is true, the probability of obtaining sample evidence at least as strongly in favor of a response time below 2 seconds as the observed evidence is 0.4%.

Since this probability is extremely small, the observed sample result would be unlikely if the true average response time were not below 2 seconds.

It is important to understand that the p-value does not mean that there is a 0.4% probability that the null hypothesis is true. Instead, it measures how compatible the observed data is with the null hypothesis.

Because the p-value is smaller than the significance level of 0.01, the evidence against the null hypothesis is considered statistically significant.

Step 5: Statistical Decision

Since:

we reject the null hypothesis.

Final Conclusion

There is sufficient statistical evidence at the 1% significance level to support the software company's claim that the average response time of the new application is less than 2 seconds.

The result is statistically significant because the p-value is much smaller than the chosen significance level. Therefore, based on the available sample evidence, the company can reasonably conclude that the application meets the claimed performance requirement.

Practical Interpretation

From a practical point of view, an average response time below 2 seconds may improve the overall user experience. Faster applications generally provide better usability, reduce waiting time, and may increase customer satisfaction. However, the company should continue monitoring the application's performance under different conditions, such as high user traffic, different network speeds, and increased server load.

Thus, the hypothesis test provides strong statistical support for the company's claim, but continuous performance monitoring is still important to ensure that the application consistently maintains the desired response time.

2. A digital marketing team launches a new advertisement and wants to know whether it increases the click-through rate. Before the campaign, the rate was 6%. After the campaign a hypothesis test produces a p-value of 0.12. What should the data science team conclude at the 5% significance level?

Scenario

A digital marketing team launches a new advertisement and wants to determine whether the advertisement increases the click-through rate of users. Before the campaign was launched, the click-through rate was 6%. After implementing the new advertisement, the team performs a hypothesis test and obtains a p-value of 0.12. The significance level selected for the analysis is 5%, or $\alpha = 0.05$.

The objective is to determine whether there is sufficient statistical evidence to conclude that the new advertisement has increased the click-through rate.

Answer

In this problem, the parameter of interest is the population click-through rate, represented by p .

The previous click-through rate was 6%, or:

The marketing team specifically wants to know whether the new advertisement increases the click-through rate. Therefore, the alternative hypothesis represents an increase above 6%.

Step 1: Formulation of Hypotheses

The hypotheses can be written as:

Null Hypothesis (H_0):

This hypothesis represents the assumption that the new advertisement does not increase the click-through rate beyond the previous value of 6%.

Alternative Hypothesis (H_1):

This represents the claim that the new advertisement increases the click-through rate.

Since the alternative hypothesis contains the symbol “>”, this is a right-tailed hypothesis test.

Step 2: Given Information

The information provided is:

- Original click-through rate = 6%

- $p\text{-value} = 0.12$
- Significance level = 0.05

The significance level of 0.05 means that the team requires sufficient evidence before rejecting the null hypothesis.

Step 3: Compare the p-value with α

The decision rule for the p-value approach is:

- Reject H_0 if $p\text{-value} \leq \alpha$.
- Fail to reject H_0 if $p\text{-value} > \alpha$.

Substituting the values:

Since the p-value is greater than the significance level, we fail to reject the null hypothesis.

Step 4: Interpretation of the p-value

A p-value of 0.12 indicates that, under the assumption that the advertisement does not truly increase the click-through rate, observing evidence at least as favorable to an increase as the sample result would have a probability of 12%.

A 12% probability is not sufficiently small when compared with the chosen 5% significance level.

Therefore, the observed data does not provide strong enough statistical evidence to reject the assumption that the new advertisement has not produced a genuine increase in the click-through rate.

Step 5: Statistical Decision

Since:

we fail to reject the null hypothesis.

It is important to use the phrase “fail to reject” rather than “accept the null hypothesis.” Failing to reject the null hypothesis does not prove that the advertisement has no effect. It simply means that the available sample data is not strong enough to demonstrate a statistically significant increase.

Final Conclusion

At the 5% significance level, there is insufficient statistical evidence to conclude that the new advertisement significantly increases the click-through rate above the previous value of 6%.

Therefore, the data science team should not claim that the new advertisement has been proven to improve the click-through rate based on the current hypothesis test.

Business Interpretation

The p-value of 0.12 does not necessarily mean that the advertisement is ineffective. There may still be a small positive improvement, but the available data does not provide enough evidence to establish that the improvement is statistically significant.

The marketing team may consider:

1. Collecting data from a larger number of users.
2. Running the advertisement campaign for a longer period.
3. Comparing the new advertisement with the old advertisement using an A/B test.
4. Analyzing the results for different customer groups.
5. Examining other performance measures such as conversion rate, engagement, and revenue.

Therefore, the current evidence is not strong enough to confirm that the advertisement has significantly increased the click-through rate.

3. A machine-learning team develops a new classification model. The old model has an accuracy of 82%, while the new model achieves 85%. A statistical test gives a p-value of 0.01. Explain what the p-value means and whether the team should conclude that the new model performs significantly better.

Scenario

A machine-learning team develops a new classification model to improve prediction performance. The existing or old model has an accuracy of 82%, while the newly developed model achieves an accuracy of 85%. A statistical test comparing the performance of the two models produces a p-value of 0.01.

The team wants to understand the meaning of the p-value and determine whether the new model can be considered significantly better than the old model.

Answer

The observed results indicate that the new model has a higher accuracy than the old model.

- Old model accuracy = 82%
- New model accuracy = 85%

Although the new model has a higher observed accuracy, an observed difference alone is not enough to conclude that the new model is genuinely better. The difference could possibly occur due to random variation in the test data.

Therefore, a hypothesis test is used to determine whether the improvement is statistically significant.

Step 1: Formulation of Hypotheses

The hypotheses can be expressed as:

Null	Hypothesis	(H ₀):
The new model does not perform better than the old model.		

Alternative	Hypothesis	(H ₁):
The new model performs better than the old model.		

In terms of accuracy, if and represent the true accuracies of the two models:

The alternative hypothesis focuses specifically on whether the new model provides an improvement.

Step 2: Given Information

The following information is available:

- Accuracy of old model = 82%
- Accuracy of new model = 85%
- p-value = 0.01

A commonly used significance level is:

Step 3: Meaning of the p-value

The p-value represents the probability of obtaining results as extreme as, or more extreme than, the observed improvement, assuming that the null hypothesis is true.

Therefore, a p-value of 0.01 means that if the new model did not actually perform better than the old model, the probability of obtaining an observed improvement of this magnitude or greater due to random sampling variation would be approximately 1%.

The small p-value indicates that the observed improvement is unlikely to be explained only by random chance.

It is important to note that the p-value does not represent the probability that the null hypothesis is true. Instead, it indicates how unusual the observed data would be if the null hypothesis were true.

Step 4: Compare the p-value with the Significance Level

Assuming a 5% significance level:

Compare:

Since the p-value is smaller than the significance level, the null hypothesis is rejected.

Step 5: Statistical Decision

The statistical decision is:

Reject the null hypothesis.

There is sufficient evidence to support the conclusion that the new model performs significantly better than the old model.

Final Conclusion

The new classification model achieves an accuracy of 85%, compared with 82% for the old model. The hypothesis test produces a p-value of 0.01, which is smaller than the commonly used 5% significance level.

Therefore, the improvement in performance is statistically significant at the 5% level.

The team can conclude that there is sufficient statistical evidence to suggest that the new machine-learning model performs better than the old model.

Practical Interpretation

Statistical significance does not automatically guarantee practical importance. Although the improvement from 82% to 85% is statistically significant, the team should also consider whether a 3-percentage-point improvement is sufficiently valuable for the intended application.

For example, the new model may require:

- More computational resources.
- Longer training time.
- Greater memory usage.
- More complex maintenance.
- Additional deployment costs.

If the 3% improvement provides meaningful benefits and the additional cost is acceptable, the new model may be deployed.

The team should also evaluate other metrics such as:

- Precision
- Recall
- F1-score
- False positive rate
- False negative rate
- Computational efficiency

Overall Conclusion

Based on the p-value of 0.01, the new model demonstrates a statistically significant improvement over the old model at the 5% significance level. Therefore, the machine-learning team has sufficient evidence to conclude that the new classification model performs significantly better.

4. A factory claims that its defect rate is 2%. After randomly inspecting products, a 95% confidence interval for the defect rate is found to be (2.4%, 3.6%). At the same time, a hypothesis test gives a p-value of 0.01. Interpret both results and explain whether the company's claim should be rejected.

Scenario

A factory claims that its defect rate is 2%. To verify this claim, a random inspection of products is conducted. Based on the sample data, a 95% confidence interval for the true defect rate is calculated as:

In addition, a hypothesis test is conducted and produces a p-value of 0.01.

The objective is to interpret both the confidence interval and the p-value and determine whether the company's claim that the defect rate is 2% should be rejected.

Answer

This problem involves two important statistical methods:

1. Confidence interval estimation
2. Hypothesis testing using a p-value

Both methods provide information about whether the company's claimed defect rate of 2% is consistent with the sample data.

Part A: Interpretation of the 95% Confidence Interval

The calculated 95% confidence interval is:

This interval provides a range of plausible values for the true population defect rate based on the sample data and the statistical method used.

The interval indicates that the estimated defect rate is between 2.4% and 3.6%.

The company's claimed value is:

When comparing the claimed defect rate with the confidence interval, we observe that:

Therefore, the claimed value of 2% is not included within the 95% confidence interval.

Furthermore, the entire confidence interval is above 2%. This suggests that the true defect rate is likely to be higher than the company's claimed value.

Thus, based on the confidence interval alone, the sample data does not support the company's claim of a 2% defect rate.

Part B: Formulation of Hypotheses

To statistically test the company's claim, the hypotheses can be formulated as:

Null Hypothesis (H_0):

The null hypothesis assumes that the true defect rate is exactly 2%.

Alternative Hypothesis (H_1):

The alternative hypothesis states that the actual defect rate is different from 2%.

Because the question asks whether the company's claim should be rejected, a two-tailed test is appropriate when testing equality against any difference.

Part C: Interpretation of the p-value

The hypothesis test gives:

For a 95% confidence level, the corresponding significance level is generally:

The decision rule is:

- If $p\text{-value} \leq 0.05$, reject H_0 .
- If $p\text{-value} > 0.05$, fail to reject H_0 .

Comparing the values:

Therefore, the null hypothesis is rejected.

The p-value of 0.01 means that, if the true defect rate were actually 2%, obtaining sample results as inconsistent with the 2% claim as those observed, or more extreme, would have a probability of approximately 1%.

Since this probability is very small, the observed data provides strong evidence against the claim that the true defect rate is 2%.

Part D: Relationship Between the Confidence Interval and p-value

The confidence interval and hypothesis test both lead to the same overall conclusion.

Confidence Interval Result:

The 95% confidence interval is:

The claimed value of 2% is not included in this interval. Therefore, the claim is inconsistent with the interval estimate.

Hypothesis Test Result:

Therefore, the null hypothesis that the defect rate is 2% is rejected.

Both methods provide evidence against the company's claim.

Final Conclusion

The company's claim that the defect rate is 2% should be rejected based on the available statistical evidence.

The 95% confidence interval indicates that the true defect rate is plausibly between 2.4% and 3.6%, and the entire interval is above 2%. This suggests that the actual defect rate is likely higher than the claimed value.

In addition, the p-value of 0.01 is less than the 5% significance level, providing statistically significant evidence against the null hypothesis.

Therefore, there is sufficient evidence to conclude that the actual defect rate is significantly different from 2% and, based on the confidence interval, is likely to be higher than 2%.

Business and Quality-Control Implication

From a business perspective, this result is important because a higher defect rate may lead to:

- Increased production costs.
- More product returns.
- Customer dissatisfaction.
- Warranty expenses.
- Damage to the company's reputation.

The company should investigate the production process to identify possible causes of the higher defect rate. Quality-control measures, equipment maintenance, employee training, and inspection procedures may need to be improved.

Therefore, the statistical results suggest that corrective action should be considered rather than continuing to rely on the original 2% defect-rate claim.

5. An online shopping company wants to determine whether a new user- interface design increases the average amount spent per customer. The data scientist calculates a 95% confidence interval for the difference in average spending as (₹120, ₹480) and obtains a p-value of 0.002. Interpret the confidence interval and p-value, and provide a business recommendation.

Scenario

An online shopping company introduces a new user-interface design and wants to determine whether the new design increases the average amount spent by each customer.

A statistical analysis produces a 95% confidence interval for the difference in average spending of (₹120, ₹480). In addition, a hypothesis test gives a p-value of 0.002.

The objective is to interpret both the confidence interval and the p-value and provide an appropriate business recommendation.

Answer

The company is interested in determining whether the new user-interface design has a positive effect on the average amount spent per customer.

Let the difference in average spending be represented as:

A positive difference indicates that the new user interface increases customer spending.

Part A: Interpretation of the 95% Confidence Interval

The calculated 95% confidence interval is:

This means that the estimated difference in average customer spending is positive and lies between ₹120 and ₹480.

Based on the sample data, the new user-interface design is estimated to increase the average amount spent per customer by at least ₹120 and potentially as much as ₹480.

The most important observation is that the entire confidence interval is above zero.

Since zero is not included in the confidence interval, the data provides evidence that there is a positive difference in average spending.

If the confidence interval had included zero, for example, (−₹50, ₹300), it would suggest that the new design might have no effect or could even reduce spending. However, this is not the case.

Because the entire interval contains positive values, the results indicate that the new user-interface design is associated with an increase in average customer spending.

Part B: Formulation of Hypotheses

The hypotheses can be stated as:

Null Hypothesis (H_0):

The new user-interface design does not increase the average amount spent per customer.

In terms of the difference in average spending:

Alternative Hypothesis (H_1):

The new user-interface design increases the average amount spent per customer.

This is a one-tailed test because the company is specifically interested in determining whether the new design produces an increase.

Part C: Interpretation of the p-value

The hypothesis test produces:

Assuming a significance level of:

we compare the p-value with the significance level:

Since the p-value is much smaller than 0.05, the null hypothesis is rejected.

A p-value of 0.002 means that if the new user-interface design actually did not increase customer spending, obtaining an increase as large as the observed result, or a more extreme increase, would have a probability of approximately 0.2%.

This is very strong evidence against the null hypothesis.

Part D: Statistical Decision

Since:

we reject the null hypothesis.

Therefore, there is sufficient statistical evidence to conclude that the new user-interface design increases the average amount spent per customer.

Part E: Combined Interpretation of Confidence Interval and p-value

Both statistical results support the same conclusion.

Confidence Interval:

The entire interval is positive, indicating that the new design is associated with an increase in average customer spending.

p-value:

The p-value is significantly smaller than the significance level, indicating that the observed increase is statistically significant.

Therefore, both the confidence interval and the hypothesis test provide strong evidence that the new user-interface design has a positive effect on customer spending.